

Estimation of relative risk, odds ratio and their logarithms with guaranteed accuracy and controlled sample size ratio

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Abstract

Given two populations from which independent binary observations are taken with parameters p_1 and p_2 respectively, estimators are proposed for the relative risk p_1/p_2 , the odds ratio $p_1(1-p_2)/(p_2(1-p_1))$ and their logarithms. The estimators guarantee that the relative mean-square error, or the mean-square error for the logarithmic versions, is less than a target value for any $p_1, p_2 \in (0, 1)$, and the ratio of average sample sizes from the two populations is close to a prescribed value. The estimators can also be used with group sampling, whereby samples are taken in batches of fixed size from the two populations. The efficiency of the estimators with respect to the Cramér–Rao bound is good, and in particular it is close to 1 for small values of the target error.

Keywords: Estimation, sequential sampling, group sampling, relative risk, odds ratio, log odds ratio, mean-square error, efficiency.

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1 Introduction

Let $p_1, p_2 \in (0, 1)$ denote the probabilities of occurrence of a given dichotomous attribute in two different populations. The problem of estimating the *relative risk* (RR) or *risk ratio*,

$$\theta = \frac{p_1}{p_2}, \quad (1)$$

from binary observations of the two populations arises frequently in medical and social sciences, as well as in other fields. For example, in a phase-III clinical trial of a vaccine ([Armitage et al, 2002](#), chapter 18) the relevant attribute is the presence of a disease, and the two populations are vaccinated and non-vaccinated people. The *odds ratio* (OR),

$$\psi = \frac{p_1(1-p_2)}{p_2(1-p_1)}, \quad (2)$$